

Table 2. False-premise correction and normal-question preservation

Method	Qwen2.5 7B				GPT-6 Luna				Gemma 4 12B				Qwen3.8 27B OFF				Qwen3.8 27B ON			
	Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE	
	FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑
Plain	2.4	100.0	—	—	54.5	91.0	72.7	98.9	4.3	100.0	—	—	49.7	79.9	—	—	40.7	88.6	—	—
Balanced instruction	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CoT	0.7	99.3	—	—	—	—	—	—	3.4	99.3	—	—	54.7	71.1	—	—	43.1	87.9	—	—
PreWoMe	—	—	—	—	88.9	59.0	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Extract + Verify	—	—	—	—	87.9	48.0	—	—	—	—	—	—	—	—	—	—	—	—	—	—
TF-IDF text gate	50.7	60.4	—	—	—	—	—	—	63.8	50.3	—	—	68.3	49.0	—	—	78.2	48.3	—	—
Direct gate	6.9	90.6	—	—	—	—	—	—	37.2	83.9	—	—	53.0	72.5	—	—	53.5	75.8	—	—
CoT gate	21.3	85.9	—	—	—	—	—	—	43.4	67.1	—	—	62.1	59.7	—	—	62.1	60.4	—	—
Probe gate	44.8	64.4	—	—	n/a	n/a	n/a	n/a	—	—	—	—	—	—	—	—	—	—	—	—
GEPA-gate	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
GEPA	—	—	—	—	78.8	72.0	74.7	98.0	—	—	—	—	—	—	—	—	—	—	—	—
GEPA (both stages)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Ours	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Stored measurements, not completed 3-fold results. Planned Cancer-Myth: grouped 3-fold OOF; CREPE: fixed train/dev/test.
Cancer-Myth: public-model entries use the historical full corpus (up to 583 FPQ / 149 NFP); Luna entries use 99 FPQ / 100 NFP.
CREPE: Luna uses 751 FPQ / 2,253 TPQ. Public-model and Luna runs use different judges; see README and result_sources.md.
Direct / CoT gate values use historical label-only routing. They do not measure the newly proposed token-score / review-passing protocol.
GEPA values are stored direct-answer test results, not fold reruns. GEPA-gate, both-stage GEPA and Ours remain unmeasured here.
All values are score >= 4 (%), from single runs. Dash = no compatible measurement; n/a = native hidden probe unavailable.